

aTOMos v3.6.1.5

× UGTS

Conjoined satnav kernel

Satellite positioning. Geometric relations. Packed state.

One conjoined computational definition.

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Engine base	aTOMos 3.6.1
Geometric base	UGTS-KC 3.6.2 / SCLP
Specialized release	3.6.1.5 / SN1
GPU target	RTX 5070 Ti Laptop / 12 GB
Implementation	C++17 + CUDA / FP64
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The conjoined calculation

Prepared satellite observations → aTOMos observation masks → weighted position/clock estimate → local UGTS geometry → log-polar state, OTAN2 and two exact 64-bit key layouts. The same mathematical functions support the CPU reference and the CUDA implementation.

Unified specification and accompanying source package. The GPU implementation includes native double reads, integer-texture double reads, chunked execution and an on-device verification matrix.

Release definition

aTOMos v3.6.1.5 is the satellite-positioning specialization of the conjoined aTOMos and UGTS engine. The aTOMos word/state contract and UGTS geometric/packing contract are joined in one input, evaluation and result sequence. The navigation equations are a new, explicitly named profile: **SN1**.

Layer	Conjoined role
aTOMos 3.6.1	ASA, NA, POPCNT, whole-word disposition, explicit JK/encoded blend, literal OTAN2 and its separate completion.
UGTS-KC 3.6.2 SCLP	Cone/sphere relations, log-polar metric and calculus, distinct topology profiles, phase/winding, bounded grammar and 20/18/14/12 key codecs.
SN1 addition	Prepared-code receiver positioning in ECEF, one clock-bias state, weighted least squares, formal uncertainty and the local ENU binding.
GPU implementation	Batched FP64 solves and geometric enrichment; global or integer-texture input path; explicit chunk-boundary continuity.
Runtime result	One plain CSV record per epoch, with numeric values, support bits, statuses, full time and compact keys. No runtime provenance-chain service.

The supplied UGTS PDF is the 19-page SCLP document whose internal content pages are numbered 1–16. The aTOMos base is the 3.6.1 master contract retained in this project. This volume restates the operative definitions together rather than concatenating the old documents or claiming that either source already contained a GNSS receiver equation.

Build and execution status

The CPU C++ implementation, mathematical tests and independent NumPy positioning oracle were executed. The CUDA implementation is delivered as **source targeting SM_120**. CUDA compilation, GPU execution and Compute Sanitizer were **not run**: the preparation environment had no nvcc or accessible NVIDIA GPU. The exact categories appear in the package's validation record.

The selected application is receive-only positioning and local geometric queries from prepared observations. The output is an estimated position/clock state and diagnostics, not a route, flight-control command or actuator message.

The one-pass conceptual flow

1 / Prepared observation epoch

Common reception-time ECEF frame, corrected code ranges, positive supplied standard deviations, initial position/clock, full GPST tick and explicit operator inputs.

2 / aTOMos selection

Slot-support word → ASA → NA → intersection count → whole-word result. J/K and encoded blend retain their own input meanings.

3 / Satellite-positioning solve

Selected code observations → iterative linearization → streaming weighted Givens QR → receiver ECEF position and clock metres.

4 / UGTS geometric evaluation

Local ENU coordinate → cone and paired-sphere residuals → log-polar chart and explicit origin-core state. The third spatial coordinate stays in the result.

5 / Packing and differential observation

Quantized (ρ, θ, X, ϕ) → contiguous and Morton keys. Consecutive chart samples → literal OTAN2 plus separately labelled directed residual.

6 / Publication

Full coordinates, measurement-fit statistics, formal sigmas, numeric support classifications, JK/blend fields, absolute time and compact addresses in one row.

The runtime has two GPU kernels: independent per-epoch positioning, then enrichment of the completed solutions. The second kernel receives the preceding chunk's last solution explicitly. A chunk size of one therefore has the same specified chart-difference semantics as a larger batch.

The UGTS sequence *support* → *compatibility* → *guard* → *verified event* → *transition* → *lineage* is retained as typed query/event logic. Here, its geometric event describes the solved *estimated point*; a supplied covariance is not automatically a deterministic position-bound certificate. [UGTS, §§14,17]

Contents

Release definition	2
The one-pass conceptual flow	3
1 The conjoined state and symbol contract	5
2 The new satnav observation model / SN1	6
3 Reference frame and time	7
4 Prepared input and aTOMos observation words	8
5 Position solution by streaming weighted QR	9
6 Residuals and formal uncertainty	10
7 JK and encoded blend inside the satnav record	11
8 UGTS finite cone and paired supports	12
9 Translational sweep and interval algebra	13
10 Local log-polar metric and kinematics	14
11 OTAN2, consecutive samples and invariant meanings	15
12 Full time, phase, winding and topology	16
13 Both exact 64-bit codecs	17
14 Radix refinement, grammar and constraint calculus	18
15 Composition and numerical event semantics	19
16 CUDA kernels and observation residency	20
17 Precision, memory and the laptop target	21
18 Recovery and positioning invariants	22
19 Retained producer and engine interfaces	23
20 The actual worked satnav output	24
21 Verification completed in this edition	25
22 Build, run and device review	26
23 Record schema and integration entry points	27
24 Package contents and conjoined coverage	28
Technical references	29

1 The conjoined state and symbol contract

Let a positioning epoch have observation set O_k , explicit operator inputs M_k , receiver state $z_k = (x_k, y_k, z_k^{\text{pos}}, b_k)$ and a diagnostic record D_k . The four solver coordinates are all measured in metres: the last is $b = c \delta t_r$, not a phase angle.

$$\mathcal{E}_{3.6.1.5} : (O_k, M_k, z_k^{(0)}, \Theta) \mapsto (\hat{z}_k, D_k). \quad (1)$$

Symbol	Meaning	Distinct from
T_c, α_c	Cone slant and half-angle	Time and reference-axis angle.
$\mathbf{r}, \mathbf{s}_i$	Receiver/satellite ECEF positions	The two-scalar log chart.
d_i	Geometric satellite range	ρ , the logarithmic local radius.
ρ, θ	UGTS local spatial chart	Independent hoop/hinge ϕ_h .
ϕ_h	UGTS hinge angle, supplied	Satellite carrier phase or clock bias.
α_r	OTAN2 reference orientation	Cone half-angle α_c .
X	Declared modular time index	Full GPST tick and clock bias.
J_{JK}, K_{JK}, q	Explicit bit inputs and state	Geometric key K_C or K_M .
P_i^*	Corrected code observation	Source phase-projection helper.
σ_i	Supplied observation deviation	One-bit jitter selector s_j .

The aTOMos spatial lower-case phase maps to **UGTS** θ in this specialization. UGTS ϕ_h stays independent. The older aTOMos spatial alias does not collapse these two UGTS fields. The fourth 12-bit key component stores the supplied hinge angle, not OTAN2 or an extra clock quantity. [UGTS, §2.1; §12][aTOMos 3.6.1 contract]

The original word core and the new numerical estimate coexist. A word disposition can prevent a solution from having enough observations, but it does not make a quantized key equal to an ECEF position. A missing result has a status; a legitimate zero clock or zero residual remains an actual value.

2 The new satnav observation model / SN1

The supplied sources define geometry, state and operators; they do not give a GNSS pseudorange model. SN1 adds the standard prepared-code positioning relation, using satellite positions and corrected observations that share a reception-time frame. [N1, N2]

$$P_i^* = \|\mathbf{s}_i - \mathbf{r}\| + b + \varepsilon_i, \quad b = c \delta t_r, \quad i \in \mathcal{A}. \quad (2)$$

Here $\mathcal{A}$ is the explicitly selected satellite-slot set, not a searched target set. The input has independent positive standard deviations σ_i and weights $w_i = 1/\sigma_i^2$. The selected state minimizes

$$J(\mathbf{r}, b) = \sum_{i \in \mathcal{A}} \frac{[P_i^* - (\|\mathbf{s}_i - \mathbf{r}\| + b)]^2}{\sigma_i^2}. \quad (3)$$

This weighted estimator is the new specialization, not a replacement for an ASA or UGTS definition. [N3]

2.1 What “corrected” means

A conventional raw code equation contains satellite clock, propagation and signal-dependent terms. This interface places the modeled nonreceiver-clock terms on the data side:

$$P_i^* = P_i + c \delta t_{s,i} - T_i - I_i - d_{\text{code},i}. \quad (4)$$

The signs and particular signal corrections must match the upstream processor’s convention. The kernel does not apply a second copy of a correction that its input has already incorporated. [N1]

The satellite location is evaluated for emission and expressed in the ECEF frame tied to reception. Those epochs are different. Earth rotation during propagation is part of the upstream coordinate preparation; a satellite location at reception is not silently substituted for its emission location. [N2]

2.2 Model dimension

There are four unknowns: three position components and one receiver-clock bias. At least four independent selected observations are required. One clock-bias family is assumed. Unresolved inter-system or inter-signal offsets need additional unknowns or upstream corrections; packing them into a common bitset does not remove the extra parameters.

The current code handles prepared code observations. It does not decode a navigation message, acquire an RF signal or resolve carrier-phase ambiguities. That makes its input contract precise and its positioning calculation usable with an upstream observation exporter.

3 Reference frame and time

The selected terrestrial frame is ECEF, with a local East–North–Up frame anchored at a supplied ellipsoidal latitude φ_0 , longitude λ_0 and height h_0 . SN1 uses the WGS84 ellipsoid constants $a = 6378137$ m and $1/f = 298.257223563$. [N4, N5]

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \quad e^2 = f(2 - f), \quad (5)$$

$$\mathbf{r}_0 = \begin{pmatrix} (N + h_0) \cos \varphi_0 \cos \lambda_0 \\ (N + h_0) \cos \varphi_0 \sin \lambda_0 \\ (N(1 - e^2) + h_0) \sin \varphi_0 \end{pmatrix}. \quad (6)$$

The local coordinate is $\mathbf{p} = R_{ENU}(\mathbf{r} - \mathbf{r}_0)$, with

$$R_{ENU} = \begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\cos \lambda_0 \sin \varphi_0 & -\sin \lambda_0 \sin \varphi_0 & \cos \varphi_0 \\ \cos \lambda_0 \cos \varphi_0 & \sin \lambda_0 \cos \varphi_0 & \sin \varphi_0 \end{pmatrix}. \quad (7)$$

The transformation is an orthonormal rotation of a translated vector. The geodetic Up vector is normal to the chosen ellipsoid at the anchor; it is not generally the geocentric radial vector. [N4]

3.1 Three dimensions are retained

The local spatial log chart uses only (E, N) , while U remains a separate metre-valued field. The chart is a view/index of a solved three-dimensional point, not an assertion that the entire point is encoded by two scalars. A change of anchor changes the local view without changing the underlying ECEF solution.

3.2 One time scale per batch

The `tick_ms` input is an absolute, monotonically increasing GPST millisecond tick under the declared common origin. It is not UTC, and no leap-second conversion is inferred by this CSV reader. The key's 14-bit time coordinate uses `tick_ms mod 16384`; the full tick is retained in every output. Repeated modular time values are therefore not mistaken for identical epochs.

The numerical clock bias b is a separate unknown. Dividing b by $c = 299792458$ m/s yields seconds of receiver offset under Eq. (2); it does not alter the epoch identifier or the hinge angle.

4 Prepared input and aTOMos observation words

The runnable profile assigns one bit to each of at most 32 observation slots in an epoch. Let $V = (1 \ll m) - 1$ for $m < 32$, or all 32 bits for $m = 32$. The support word is $X = V$ because every present record has a declared slot; quality selection is supplied by the ASA/NA masks.

$$A = X \wedge M_{\text{ASA}} \wedge V, \quad N = A \wedge M_{\text{NA}}, \quad (8)$$

$$H = \text{popcount}(N \wedge M_{\partial}), \quad Y = \begin{cases} 0, & H > 0, \\ N, & H = 0. \end{cases} \quad (9)$$

The active satellite set is $\mathcal{A} = \{i : Y_i = 1\}$. This is the retained whole-word rule, applied to a new *observation-slot dictionary*. [aTOMos 3.6.1 contract][SN1]

4.1 Exact semantic preservation

A boundary intersection clears the *whole selected word*; it does not merely clear the intersecting bit. A user-supplied observation exclusion therefore belongs in the ASA or NA mask when it is intended to remove only that observation. The implementation does not silently change the boundary operator to make a GNSS quality-control algorithm out of it.

Epochs with $H > 0$ return `whole_word_absorbed`. Those with fewer than four surviving observations return `insufficient_observations`. More than four slots does not guarantee rank; the solver separately checks the geometry's numerical rank.

4.2 CSV binding

An epoch file supplies full time, initial four-state estimate, hinge angle, masks, J/K bits and the three-bit blend encoding. An observation file supplies epoch ID, contiguous slot, unique satellite ID, three satellite coordinates, corrected code and sigma. A row referencing an absent epoch, duplicate slot, nonfinite number or a nonpositive sigma is an input error, not a missing observation silently set to zero.

The files represent prepared, receive-only observations. There is no hidden radio configuration, antenna command or route generator in the parser. The absent physical frontend from the aTOMos 3.6.1 source-profile obligations is not claimed to have been completed merely by adding a CSV interface.

5 Position solution by streaming weighted QR

At estimate $(\mathbf{r}^{(k)}, b^{(k)})$, write

$$d_i^{(k)} = \|\mathbf{r}^{(k)} - \mathbf{s}_i\|, \quad v_i^{(k)} = P_i^* - d_i^{(k)} - b^{(k)}, \quad (10)$$

$$g_i^{(k)} = \left[\frac{x^{(k)} - s_{ix}}{d_i^{(k)}}, \frac{y^{(k)} - s_{iy}}{d_i^{(k)}}, \frac{z^{(k)} - s_{iz}}{d_i^{(k)}}, 1 \right]. \quad (11)$$

The whitened row and right-hand side are $a_i = g_i/\sigma_i$ and $\tilde{v}_i = v_i/\sigma_i$. The update minimizes $\|A\Delta z - \tilde{v}\|_2^2$. [N1, N3]

5.1 Bounded per-thread workspace

The implementation inserts each selected row into an upper-triangular 4×4 factor R and a four-element transformed right-hand side. For a diagonal element r_{kk} and incoming component a_k ,

$$\gamma = \text{hypot}(r_{kk}, a_k), \quad c = r_{kk}/\gamma, \quad s = a_k/\gamma. \quad (12)$$

The Givens rotation replaces the two rows so the new incoming component is zero. The same rotation is applied to the right-hand side. A zero pair is skipped. Repeating this for all selected rows produces the least-squares triangular system without forming normal equations for the update.

Back substitution solves $R\Delta z = z_R$, then the four-state estimate is updated. The loop stops when both the position-increment norm and the clock-increment magnitude are no greater than 10^{-4} m. The profile uses at most 20 iterations; exceeding this returns a distinct `iteration_limit` status.

5.2 Rank and finite arithmetic

A factor is treated as numerically rank deficient if a diagonal magnitude is no more than 10^{-11} times the largest diagonal, or if a required factor is nonfinite. The four-state units are balanced because the clock unknown is in metres. This diagonal test is a specified numerical criterion, not a rank-revealing pivot theorem for every possible geometry.

Satellite coordinates and corrected observations remain fixed during one solve. Only the receiver state is iterated. The coordinate-preparation convention in Section 3 therefore matters; an unmodeled change of satellite epoch is not repaired by extra iterations.

6 Residuals and formal uncertainty

After the final iteration, the implementation rebuilds the whitened geometry at the returned state. It reports

$$\text{RMS} = \sqrt{\frac{1}{m} \sum_i v_i^2}, \quad \chi^2 = \sum_i (v_i / \sigma_i)^2. \quad (13)$$

The latter is a normalized residual sum; no unknown global noise scale is silently substituted for the input sigmas. With four fitted parameters the nominal residual degrees of freedom are $m - 4$ when the selected linearized geometry is full rank.

For independent observation errors with the supplied variances, the formal linearized covariance is

$$P = (A^T A)^{-1} = R^{-1} R^{-T}. \quad (14)$$

The code solves four triangular systems to construct its diagonal and reports the four formal standard deviations. These values refer to the stated observation model, chosen weights and local linearization; they are not deterministic support radii or a claim that all real errors were modeled. [N3]

6.1 No implicit robust reweighting

The baseline does not discard a large residual automatically, perform subset solution separation or invent new observation sigmas. The supplied ASA/NA masks and sigmas are explicit inputs. A new robust-estimation profile would identify its own estimator and tests instead of altering the result under the same profile tag.

6.2 Positioning result states

State	Meaning
ok	Converged four-state solution with finite full-rank final geometry.
insufficient_observations	Fewer than four selected observations.
whole_word_absorbed	The retained boundary rule clears the selected word.
rank_deficient	Geometry fails the stated numerical rank criterion.
iteration_limit	Finite estimate returned without meeting convergence thresholds.
invalid_input / numeric_failure	Required types/domains or finite arithmetic fail.

Only ok results enter the geometric enrichment path. Unavailable floating fields are serialized as nan together with an explicit status, not misrepresented by a valid-looking zero coordinate.

7 JK and encoded blend inside the satnav record

The aTOMos sequential bit operator remains

$$q^+ = (J \wedge \neg q) \vee (\neg K \wedge q), \quad q, J, K \in \{0, 1\}. \quad (15)$$

J	K	q^+	Meaning
0	0	q	Hold
1	0	1	Set
0	1	0	Reset
1	1	$1 - q$	Toggle

The prepared epoch supplies its own q_{before} , j and k . Batch positioning is parallel across epochs and does not reconstruct an unknown control history. A stateful caller can supply successive before-states; the CSV explicitly states each epoch's bit inputs. [aTOMos 3.6.1 contract]

The encoded circle-plus result remains

$$s_{\text{blend}} = n_b \oplus a_b \oplus \beta_b. \quad (16)$$

The file stores these three bits in the integer `blend_bits`; its high-to-low bits are north, axis and kinematic. Neither this parity nor the JK state is inferred from a position error or an OTAN2 residual.

7.1 Retained independence

A solver can return a nonzero JK state while its observation word is absorbed. Those are different results with different operands. Likewise, a key can be unavailable because a valid position lies outside the local chart, while the ECEF solution itself remains valid.

The broader aTOMos alignment, sweep-limit and active-fringe predicates remain independently typed roles. The selected SN1 executable uses explicit observation masks rather than a fabricated beam-alignment classifier. Its geometric classes annotate the estimated receiver point. No satellite-navigation number is automatically recast as a set/reset command or a source-defined physical fringe measurement.

7.2 Nucleus and state

The earlier engine's identity/state distinction remains conceptually compatible: a receiver/session identifier is not a coordinate or a mask. In the selected plain-record runtime, epoch IDs and full time supply the needed association without adding a content-addressed ledger. The author attribution is carried in this edition's identity fields, not used as an executable permission token.

8 UGTS finite cone and paired supports

The conjoined geometry retains the finite right circular cone defined by slant $T_c > 0$ and half-angle $0 < \alpha_c < \pi/2$:

$$h = T_c \cos \alpha_c, \quad R = T_c \sin \alpha_c. \quad (17)$$

For query $\mathbf{p}$, apex $\mathbf{c}$ and unit axis $\mathbf{a}$, use

$$z = \mathbf{a} \cdot (\mathbf{p} - \mathbf{c}), \quad q = \|(\mathbf{p} - \mathbf{c}) - z\mathbf{a}\|. \quad (18)$$

The Euclidean signed distance is the signed distance of (q, z) to the filled meridian triangle $(0, 0), (R, h), (-R, h)$, negative for the cone interior. The implementation takes the minimum distance to the three boundary segments and assigns the sign using $0 \leq z \leq h$ and $qh \leq zR$. This realizes the exact primitive defined in the supplied document. [UGTS, §3.1; Figure 2 on PDF p.5]

The two spherical relations are separately evaluated:

$$f_L(\mathbf{p}) = \|\mathbf{p} - \mathbf{c}_L\| - R_L, \quad f_R(\mathbf{p}) = \|\mathbf{p} - \mathbf{c}_R\| - R_R. \quad (19)$$

The two-bit output retains membership in each sphere and their overlap. It does not collapse two independent supports into a biological or sensor claim. [UGTS, §3.2]

8.1 SN1 local-support example

The executable uses a declared local example: cone apex $(0, 0, -100)$ m, Up axis, slant 1000 m and half-angle $\pi/3$; spheres centered at $(0, 0, 0)$ and $(200, 0, 0)$ m with radius 500 m. The point-relation margin is 0.001 m. These are editable **SN1 fixture values**, not numerical antenna parameters found in either source.

The relation class is interior when the residual is below $-\epsilon_g$, exterior above $+\epsilon_g$, and margin-band otherwise. The underlying residual remains in the output. This evaluates a geometric relation at the estimated point; its class does not feed back into the position solve or alter J/K inputs.

8.2 One exact check

For $T_c = 2$, $\alpha_c = \pi/6$, zero apex, Up axis and $\mathbf{p} = (0.15, 0, 0.6)$, the side residual is $0.15 \cos(\pi/6) - 0.6 \sin(\pi/6) = -0.1700961894\dots$. This reproduces the source's displayed interior value with explicitly stated query parameters and is checked in the C++ tests.

9 Translational sweep and interval algebra

For a fixed-orientation cone and finite translation $s(u)$, the UGTS sweep relation is

$$F(\mathbf{p}) = \inf_{u \in [0,1]} d_C(\mathbf{p} - s(u)). \quad (20)$$

For a linear path of length L and $n \geq 2$ equally spaced samples, let m_n be the minimum sampled value. The retained interval is

$$m_n - \frac{L}{2(n-1)} \leq F(\mathbf{p}) \leq m_n. \quad (21)$$

The closest continuous parameter lies within half a sample step of some sampled parameter, and signed distance under translation is 1-Lipschitz. This gives the bound in Eq. (21). [UGTS, §4]

A zero-length sweep has zero interval width. The radius $L/[2(n-1)]$ decreases as n grows. The numerical intervals from *arbitrary nonnested* sample sets need not themselves nest. Nested sampling or intersection with previously established intervals supplies a nested certificate sequence. This distinction makes the source's sample-density language explicit rather than silently strengthening the theorem.

Rotating cones require additional information beyond this translation proof. The conjoined helper implements the stated fixed-orientation translational relation, not an unprovided general rotational envelope.

9.1 The two jitter intervals are different objects

UGTS defines $f_j = f + \epsilon_j \sigma$, $\sigma \in \{-1, +1\}$. Its page-5 display equates intervals centered on f_j and f . In general those intervals are not equal. This edition retains their distinct meanings:

$$I_0 = [f - \epsilon_j, f + \epsilon_j], \quad I_j = [f_j - \epsilon_j, f_j + \epsilon_j], \quad f = f_j - \epsilon_j \sigma. \quad (22)$$

I_0 is a nominal uncertainty interval; I_j contains the unperturbed f when only the signed bounded modulation is known. For $\epsilon_j > 0$, their centers differ. The independent helper and regression test preserve this distinction explicitly. [UGTS, §6]

SN1 does not add jitter to code observations. Reported measurement sigma, numerical relation margin and an optional synthetic field-jitter amplitude are three separate inputs with different meanings. Removing their type distinction would change the model rather than simplify it.

10 Local log-polar metric and kinematics

The conjoined local chart uses

$$r = \sqrt{E^2 + N^2}, \quad \rho = \ln(r/r_0), \quad \theta = \text{atan2}(N, E). \quad (23)$$

SN1 chooses $r_0 = 10,000$ m and an explicit origin core $r < 0.01$ m. *Theta is east-zero and counterclockwise in the EN plane*; it is not a north-zero navigation bearing. Up remains separately recorded. For packing, ρ must also lie in $[-20, 0]$; no clamp silently turns an out-of-range solution into a valid key. [SN1]

The retained UGTS metric, exact finite radial step and Jacobian are

$$ds^2 = r^2(d\rho^2 + d\theta^2), \quad \Delta r = r \expm1(\Delta\rho), \quad (24)$$

$$J = r \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (25)$$

The approximation $r\Delta\rho$ is only the first-order expansion of the exact radial change. The origin is an explicit core case rather than a direction manufactured by $\ln 0$. [UGTS, §5]

For supplied coordinate rates,

$$\mathbf{v} = r(\dot{\rho} \mathbf{e}_r + \dot{\theta} \mathbf{e}_\theta), \quad (26)$$

$$\mathbf{a} = r[(\ddot{\rho} + \dot{\rho}^2 - \dot{\theta}^2)\mathbf{e}_r + (\ddot{\theta} + 2\dot{\rho}\dot{\theta})\mathbf{e}_\theta]. \quad (27)$$

These formulas are retained and tested in the independent geometry helpers. The code-only positioning solver does not infer a physical acceleration model or claim a velocity estimate from a single epoch.

10.1 Gradient and field versus coordinate effect

For chart derivatives (f_ρ, f_θ) , $\nabla f = (f_\rho \mathbf{e}_r + f_\theta \mathbf{e}_\theta)/r$. This is a coordinate transformation. It does not automatically turn a physical force into $e^{-\rho}$ times some other vector. A mechanics model remains a separate definition, as UGTS explicitly states.

The local chart serves packing, differential observation and query geometry. The global receiver coordinates remain the three ECEF values computed from the observations. No Klein wrap or grammar step changes those coordinates during the satnav solve.

11 OTAN2, consecutive samples and invariant meanings

The literal operator retained from aTOMos is

$$\text{otan2}_s(\Delta\theta, \Delta\rho) = \arctan(\Delta\theta/\Delta\rho), \quad \Delta\rho \neq 0, \quad (28)$$

$$e_s = \text{otan2}_s - \alpha_r. \quad (29)$$

The output stores the source orientation and raw relative residual. It also stores the separately named completion $e_d = \text{wrap}_{2\pi}(\text{atan2}(\Delta\theta, \Delta\rho) - \alpha_r)$. Neither quantity is a position estimate, satellite code phase or automatically an arrival direction. [aTOMos 3.6.1 contract]

The executable uses consecutive solved epochs in one fixed local chart. Its declared increment policy is

$$\Delta\rho = \rho_k - \rho_{k-1}, \quad \Delta\theta = \text{wrap}_{2\pi}(\theta_k - \theta_{k-1}). \quad (30)$$

This is a *principal-difference* profile. It is not recovery of arbitrary physical winding from two samples. A failed solve or out-of-chart previous sample leaves the differential observation unavailable.

Law	Transformation	Preserved / expected result
R	Common radial scaling about one origin	Same $\Delta\rho$ and orientation.
W	Same positive interval for both rates	Common divisor cancels.
P	Same shift of consistently lifted phases	Angular difference unchanged.
S	Positive scaling of the increment vector	Ratio and directed angle unchanged.
F	Rotate increment and reference together	Source line class modulo π ; completion modulo 2π .
V	Reverse both components	Source unchanged; completion differs by a half-turn.

These inherited identities remain mathematical diagnostics. The satnav kernel does not turn their pass/fail state into automatic measurement rejection or J/K feedback. Its C++ tests check source/line-class behavior and positional invariances; the full inherited six-card UI is not required to run a batch.

Changing only α_r changes e_s by the negative of that change. That covariance is intentional and leaves the GNSS estimate unchanged. It is not an invariant that the engine should force to zero.

12 Full time, phase, winding and topology

UGTS keeps linear time and defines a phase clock with reference X_0 and period P :

$$u = (X - X_0)/P, \quad w = \lfloor u \rfloor, \quad \psi = u - w \in [0, 1). \quad (31)$$

Winding w distinguishes repeated phase positions. Its source example uses reference 1135, tick 1920 and period 256, producing $w = 3$ and $\psi = 0.06640625$. The units belong to that profile and are not universal satellite-navigation constants. [UGTS, §7]

SN1 retains the full absolute millisecond tick and packs only its remainder modulo 2^{14} . A modular key can repeat after 16.384 s with this tick unit; the absolute epoch does not repeat. The compact word is therefore an address component, not a complete time history.

12.1 Two distinct topology profiles

On an odd radial wrap the source half-turn profile is

$$(\rho, \theta, \phi_h, o) \mapsto (\rho', \theta + \pi, -\phi_h, -o). \quad (32)$$

The reflective Klein profile instead uses

$$(\rho, \theta, \phi_h, o) \mapsto (\rho', \pi - \theta, -\phi_h, -o). \quad (33)$$

The first map rotates the base angular circle; the second reverses its orientation. Both are preserved with separate tags. The reflective tangent map also negates angular and hinge tangent components. A chart transition is not a physical torque impulse. [UGTS, §8; Figure 4 on PDF p.8]

12.2 Selection in this specialization

The live positioning/enrichment path uses a bounded local annular chart. It does *not* apply either topological wrap to global ECEF coordinates or fold a receiver to a different Earth location. The two maps remain executable mathematical helpers for the retained UGTS state relations.

The source's half-open wrapping interval and the inclusive endpoint quantizer are distinct contracts. Wrapping $\rho = 0$ in $[-20, 0)$ produces a wrap event; quantizing $\rho = 0$ in the source's closed key range produces the maximum radial index. The profile selection determines which operation is being performed.

13 Both exact 64-bit codecs

UGTS supplies two different encodings of the same finite tuple. They are retained exactly rather than reconciled into one undocumented layout. [UGTS, §12; Table 1 and Figure 6 on PDF p.11]

Field	Bits	States	Contiguous bit range
q_ρ	20	1,048,576	63:44
q_θ	18	262,144	43:26
q_X	14	16,384	25:12
q_{ϕ_h}	12	4,096	11:0

The contiguous code is

$$K_C = (q_\rho \ll 44) \vee (q_\theta \ll 26) \vee (q_X \ll 12) \vee q_{\phi_h}. \quad (34)$$

The Morton code appends source bits in the MSB round-robin order

$$\rho_{19}, \theta_{17}, X_{13}, \phi_{11}, \rho_{18}, \theta_{16}, X_{12}, \phi_{10}, \dots \quad (35)$$

until all 64 bits have been emitted. Exhausted fields are skipped. The shared C++ and independent Python implementations each supply their own pack/unpack operations.

13.1 Exact source regression

For the source tuple (949111, 0, 1920, 227), both implementations reproduce

```
contiguous = 0xe7b77000007800e3
morton     = 0x88823bb88099128b
```

Both decode to the same tuple; the different words are intentional. Runtime CSV keys are emitted as decimal uint64 integers and should not be round-tripped through a floating-point spreadsheet cell.

13.2 Quantization policy

The radial field uses nearest endpoint-inclusive quantization over $[-20, 0]$, with half-up tie handling. Theta and hinge use periodic floor bins, and time is modular. The source gives widths/ranges but not every rounding decision; these selected rules are explicit SN1 completion choices. Continuous receiver position, Up, uncertainty and absolute time remain outside this key.

The capacity is exactly 2^{64} field combinations. This is neither a 2^{64} -entry allocation nor a promise that a finite word retains arbitrary exact real-valued state.

14 Radix refinement, grammar and constraint calculus

A key prefix is a radix-trie address. Appending b gives $p' = (p \ll 1) \vee b$ and fixes the next bit in the selected schedule. The corresponding integer-field interval narrows; a decoded physical coordinate is not necessarily halved. The independent prefix helper checks that no field interval widens after refinement. [UGTS, §11]

The retained bounded binary grammar has the schematic production

$$F(T_c) \Rightarrow F(T_c/2) [\pm \phi_h F(T_c/2)] J(\Delta r). \quad (36)$$

Its depth-4 symbolic example has 16 forward symbols and 76 total tokens. The code checks explicit finite depth/symbol capacity and balances the brackets. The turn sign is affected by the declared chirality profile. A supplied 12-bit grammar-state code and the 12-bit hinge quantizer are separate objects; this implementation does not invent the absent codebook mapping a grammar to the source's state word 1058. [UGTS, §10]

14.1 Hinge and missing shackle

The hinge state is $(\phi_h, \dot{\phi}_h, \ddot{\phi}_h)$. A mechanical model can supply coefficients for $\tau = I\ddot{\phi}_h + c\dot{\phi}_h + k\phi_h$; without those parameters, acceleration is not renamed torque. For a supplied velocity constraint $A(q)\dot{q} = 0$, deleting a row gives the exact freedom change

$$\Delta d = \text{nullity}(A') - \text{nullity}(A). \quad (37)$$

For the two independent coordinate rows in the source example, removal raises nullity from one to two. Removing a redundant row need not change nullity. Both cases are tested independently. [UGTS, §9]

For a regular zero-surface normal $\mathbf{n}$, the tangent projector is $P_T = I - \mathbf{n}\mathbf{n}^T$ when $\|\mathbf{n}\| = 1$. It preserves first-order tangency, not exact finite-step surface membership without further integration/projection.

These retained UGTS mechanisms are part of the conjoined mathematical package. They do not generate satellite ephemerides, commanded routes or an unprovided physical motion model. The SN1 positioning path uses measurements, not a grammar expansion, to determine receiver coordinates.

15 Composition and numerical event semantics

For each epoch, the conjoined state transition is

$$(O_k, M_k, q_k, z_k^{(0)}, \Theta) \mapsto (\hat{z}_k, q_k^+, \mathcal{R}_k), \quad (38)$$

where $\mathcal{R}_k$ includes every core, solve and geometric diagnostic field. The input objects are not modified. A solver failure is a result state, while an invalid file schema fails input construction before a batch is executed.

Stage	Reads	Writes / publishes
A	Slot dictionary and explicit masks	ASA, NA, hit count and output support.
B	Selected observations and initial state	Position/clock, fit statistics, formal sigmas and solve status.
C	Supplied q, J, K and blend bits	Independent bit outputs.
D	Solved ECEF point and fixed local profile	ENU point, cone/sphere relations and chart status.
E	Chart, full tick and independent hinge	Two packed keys and temporal sidecar.
F	Current and prior valid chart samples	Delta chart values and both labelled OTAN2 residual views.

15.1 The UGTS event boundary

A margin-band relation remains a band; its sign is not forced to a binary inside/outside class to satisfy a bit-only data path. The raw relation and margin classification remain alongside any membership bits. Uncertainty in a GNSS observation and uncertainty of a geometric interval remain separately typed.

The conjoined solution has no rule deriving observation masks from its own geometric support result. Such a feedback rule would change the estimator and require an explicit iteration/policy definition. The present output reports the support query after solving the measurement equation.

15.2 No compulsory provenance subsystem

This runtime emits ordinary input/output records, full ticks and source-independent epoch identifiers. It contains no mandatory hash-chain, identity gate, remote activation or provenance daemon. The retained UGTS lineage role is represented by explicit epoch succession and optional helper state, not by requiring a new cryptographic service to calculate a position.

16 CUDA kernels and observation residency

The shared implementation is in `include/atomos/nav.hpp`. It is compiled as ordinary C++ for the CPU and as host/device code for CUDA. The GPU backend contains a positioning kernel and an enrichment kernel. One CUDA thread solves one independent epoch, with up to 32 observations and a small local QR workspace.

The GPU observation representation is a structure of arrays with five components: $s_x, s_y, s_z, P^*, \sigma$. For chunk size N , the linear element address is

$$\text{index}(c, i, k) = (32c + i)N + k. \quad (39)$$

At a fixed satellite slot and component, neighboring threads read neighboring epochs. Empty trailing slots are initialized but never selected beyond the declared count.

16.1 Two input-read implementations

The global path directly loads binary64 values. The texture path binds the same bytes to an integer-pair linear texture resource, reads `int2`, then reconstructs the double with `__hioint2double(high, low)`. This preserves all 64 input bits instead of converting Earth-scale ECEF positions to float32. The texture descriptor uses element-type reads; integer fetches do not interpolate coordinates or observation values. [H4]

The texture is a read interface for data. Its creation does not pin all inputs in a hardware cache, and it does not place native instructions in a texture. All read paths execute the same QR and enrichment definitions.

16.2 Kernel boundaries

The solve kernel completes before enrichment in the same CUDA stream. Enrichment accesses the preceding result within the chunk, or receives the previous chunk's final solution and metadata by value. This explicit carry prevents a false reset of OTAN2 at every chunk boundary.

All CUDA allocation, copy, launch and synchronization results are checked. A requested GPU run does not silently substitute CPU work while reporting a GPU backend. The supplied device-review script compares both read paths and chunk sizes 1, 7 and 8192 against the independent reference.

17 Precision, memory and the laptop target

The requested target is the **RTX 5070 Ti Laptop GPU with 12 GB GDDR7**, not the desktop card's memory configuration. NVIDIA's compute-capability table lists the RTX 5070 Ti family at 12.0; the build requests native `sm_120` plus virtual `compute_120`. CUDA 12.8 introduced compiler support for SM_120. [H1–H3]

The current host ABI measures

Record	Bytes
Epoch metadata	88
Solution	112
Diagnostic	136
One observation	40

With five padded component planes for 32 observations, device payload per epoch is

$$5 \cdot 32 \cdot 8 + 88 + 112 + 136 = 1616 \text{ bytes.} \quad (40)$$

The default chunk of 8192 therefore requires 13,238,272 bytes of these buffers (12.625 MiB). Host data, temporary packed copies and runtime overhead are additional. The full 2^{64} key domain is never allocated as a dense table.

17.1 Allocation controls

The selected chunk is bounded by the user-requested size, remaining epochs, explicit byte budget, 60% of currently free device memory, and the texture element-index limit where applicable. Defaults are 8192 epochs and a 1024 MiB ceiling. Actual allocation failures still return errors if availability changes. No workload is padded to 12 GB solely because the laptop advertises that capacity.

17.2 What can benefit from batching

The intended parallel work is many independent recorded epochs or receiver-solution cases. One small receiver epoch may cost more to upload and launch than to solve on the CPU. The implementation supplies measurements rather than assuming a GPU speedup from the device name. No Tensor Core acceleration or per-clock cache throughput claim is made for this FP64 QR code.

GPU event timings cover the two kernels after one warm-up solve, excluding packing and transfer. CPU timing covers the solve/enrichment loops. Neither is an interchangeable end-to-end benchmark.

18 Recovery and positioning invariants

Theorem 18.1 (aTOMos projection). *Projection of the conjoined computation onto its ASA, NA, hit count and output-mask fields gives the original word rule on the declared observation dictionary. Projection onto JK and encoded blend gives the original bit equations on their explicit inputs.*

Proof. Those calculations use exactly the retained operators with the supplied mask and bit operands. The numerical positioning and UGTS enrichment stages write separate fields and do not change the operands. Dropping those added fields therefore recovers the original results. $\square$

Proposition 18.2 (Clock-shift relation). *Adding a common Δb to every prepared code observation leaves the position minimizer unchanged and increases the receiver-clock minimizer by Δb , subject to the same solution branch and convergence.*

Proof. Replacing P_i^* by $P_i^* + \Delta b$ and b by $b + \Delta b$ leaves every residual in the objective unchanged. Thus the objective is translated only in its clock coordinate. The supplied regression checks this relation. $\square$

Proposition 18.3 (Common weight-scale relation). *Multiplying all observation sigmas by $c > 0$ preserves the minimizing state and multiplies its formal standard deviations by c under the same linearized geometry.*

Proof. The objective is multiplied by $1/c^2$, which preserves its minimizer. The whitened matrix is divided by c , so its information matrix is divided by c^2 and covariance multiplied by c^2 . $\square$

Observation permutation, with slot masks remapped consistently, preserves the mathematical objective. Finite QR accumulation can introduce round-off differences, so the cross-implementation comparison uses a stated numerical tolerance rather than claiming identical floating-point evaluation order.

18.1 Chart/query isolation

Changing only the local anchor, support region or OTAN2 reference must not change the computed ECEF position/clock. Changing a query profile can change the local result and packed key, but not the upstream measurements. The code separates these inputs, and the tests check the no-mutation property.

19 Retained producer and engine interfaces

The satnav specialization is one named application of the total aTOMos engine. Its prepared-observation producer replaces neither the original source grammar nor the general numerical-field interface. The retained literal word-merge alternatives remain

$$\Psi_{\oplus} = ((\Psi \ll 1) \bmod 2^b) \oplus (\Psi \gg 1), \quad \Psi_{\vee} = ((\Psi \ll 1) \bmod 2^b) \vee (\Psi \gg 1). \quad (41)$$

The two-child lineage convention $\ell_b = 2\ell + b$ is a separate record-address rule; it is not the same operation as shifting a state word right. These upstream producer definitions do not calculate satellite positions from identifiers or replace observed ranges. [aTOMos 3.6.1 contract]

For a supplied field $f(t, z; \Theta)$, the generic RK4 interface remains

$$k_1 = f(t, z), \quad k_2 = f(t + h/2, z + hk_1/2), \quad (42)$$

$$k_3 = f(t + h/2, z + hk_2/2), \quad k_4 = f(t + h, z + hk_3), \quad (43)$$

$$z^+ = z + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (44)$$

No wave field, orbit propagator or receiver-motion prior is implicitly chosen by writing this numerical step. SN1 uses the prepared satellite coordinates and independent epoch solves specified earlier, not an unstated RK4 satellite-orbit implementation.

19.1 Original interfaces versus newly completed ones

The new satellite-observation equation fills one clearly defined numerical application layer. It does not fill the earlier aTOMos physical beamforming/eigenmatrix obligations, nor the UGTS optional inertia/damping model. Their types remain compatible with a future explicitly supplied implementation.

The conjoined package implements the satnav solver, masks, geometric enrichment, codecs and selected independent UGTS helpers. The original tape-universality profile, complete content-addressed definition registry and earlier panel UI are not necessary runtime dependencies for this positioning kernel. They are not claimed to have become live subsystems by renaming this release.

The full fixed-width key is still finite. A 12-bit codebook index cannot reconstruct an arbitrary continuous matrix without a declared codebook, and a one-bit support value does not retain a full residual or uncertainty. These are retained information distinctions, not disabled execution features. [UGTS, §13]

20 The actual worked satnav output

The package supplies two deterministic synthetic batches. Their satellite coordinates and corrected code observations are generated directly from the stated geometric model; they are not samples of a broadcast RF signal or a fetched real constellation.

The clean batch has 128 one-second epochs with 12 supplied satellites per epoch, except explicit mask examples selecting ten. The local reference starts near $(E, N, U) = (150, -75, 12)$ m, and the clock bias starts at 23,000 m. Initial estimates have known offsets so the nonlinear iteration is exercised.

First clean epoch field	Observed value
Selected observations	12
Solver iterations	3
Local East / North / Up	Approximately $(150, -75, 12)$ m
Clock state	Approximately 23,000 m
Source log radius ρ	-4.0881333022
Spatial theta	-0.4636476090 rad
Cone residual	-13.1422960696 m
Paired-sphere bits	3: both relations inside
Contiguous key	14676103987472629988
Morton key	14717998450411357183
OTAN2	No previous sample for epoch zero

At the next clean epoch, the reported chart increments are approximately $\Delta\rho = 0.0004283883354$ and $\Delta\theta = 0.0018122165142$. The source orientation is about 1.338668318 rad, and its residual relative to 15° is 1.076868930 rad. These values derive from the solved local positions, not a substituted map heading.

20.1 Noisy and edge batch

The second batch contains 192 epochs and deterministic code perturbations with amplitude 1.5 m. It includes one whole-word-absorbed case, one insufficient-observation case and one rank-deficient geometry. The remaining 189 epochs solve. Relative to the known synthetic positions, their 3D position RMSE is 1.716936641 m and maximum error 4.424750062 m in the executed CPU run.

The clean batch's maximum error is about 7.95×10^{-9} m because it is a noiseless, internally consistent arithmetic fixture. That figure measures numerical round-off in the example, not real GNSS measurement accuracy.

21 Verification completed in this edition

The following categories were executed using the accompanying code. Earlier UGTS test counts are not reused as though the original UGTS source archive had been supplied and rerun.

Executed check	Recorded outcome
C++ core/math suite	26 groups; 217,707 assertions passed.
Independent Python UGTS helpers	15 tests passed.
CPU CTest	Two tests passed, including the C++ suite.
CPU AddressSanitizer + UBSan CTest	The same two tests passed in the instrumented build.
Independent position oracle	320 epochs checked; 4,462 comparisons passed; 317 expected solution epochs.
Clean maximum state difference versus oracle	$3.725290298 \times 10^{-9}$ m across the four metre-valued state fields.
Noisy maximum state difference versus oracle	$4.212779459 \times 10^{-9}$ m.
CUDA compilation / execution	not_run: no nvcc or accessible NVIDIA GPU in preparation.
Compute Sanitizer / Windows build	not_run.

The independent positioning reference uses NumPy's least-squares solver, not the shared C++ streaming Givens implementation. It reconstructs the selected observations, solves the nonlinear equation independently, checks fit statistics and formal covariance, converts ENU and verifies both packed key integers.

The C++ suite additionally checks the exact UGTS source key pair, random codec round trips, cone geometry, Lipschitz behavior, sweep bounds, singular solver states, clock/weight invariances, the entire JK table, whole-word semantics and preserved height. Python tests cover prefix refinement, time/winding, both topology profiles, chart kinematics, the tangent projector, constraint nullity, grammar counts and the distinct jitter intervals.

21.1 The GPU check that remains to execute

The on-device matrix is global/texture reads $\times$ chunk sizes $1/7/8192 \times$ both fixtures: twelve runs. Every GPU result is compared with the independent oracle and CPU enrichment, including chunk boundaries. The optional memory check uses NVIDIA Compute Sanitizer on the texture path. Missing requested tooling returns an explicit unavailable category rather than a substituted success.

22 Build, run and device review

The CPU runtime needs a C++17 compiler and CMake 3.24 or newer. The CUDA configuration needs Toolkit 12.8 or newer, a host compiler supported by that toolkit, and a compatible installed NVIDIA driver. The CMake target is SM_120 plus its virtual representation. [H2, H3]

22.1 CPU reference

```
cmake -S . -B build_cpu -DCMAKE_BUILD_TYPE=Release
cmake --build build_cpu --config Release --parallel 2
ctest --test-dir build_cpu -C Release --output-on-failure
./build_cpu/atomos_satnav \
  --epochs examples/clean/epochs.csv \
  --observations examples/clean/observations.csv \
  --out clean_run.csv
```

22.2 Requested laptop GPU

```
cmake -S . -B build_gpu -DAO_ENABLE_CUDA=ON \
  -DCMAKE_BUILD_TYPE=Release \
  -DCMAKE_CUDA_ARCHITECTURES="120-real;120-virtual"
cmake --build build_gpu --config Release --parallel 2
./build_gpu/atomos_satnav --probe
./build_gpu/atomos_satnav \
  --epochs examples/clean/epochs.csv \
  --observations examples/clean/observations.csv \
  --out gpu_run.csv --backend cuda --read texture
```

On a Windows multi-configuration build, use this executable path from the supported compiler prompt:

```
build_gpu/Release/atomos_satnav.exe
```

The program's `-help` lists the anchor, axis, chunk and memory-budget options.

22.3 Full local review

```
python tools/validate.py
python tools/validate.py --gpu
python tools/validate.py --gpu --memcheck
```

The independent positioning checks need Python 3.10+ and NumPy. The pure Python geometry tests use only the standard library. Each review writes its actual outputs to a new directory. There is no background service, activation request or external data connection in the executable.

23 Record schema and integration entry points

The executable consumes two files and emits one file. Their exact headers and column semantics are specified in docs/INPUT.md. Plain CSV keeps the receive-only calculation usable without requiring a runtime database, hash chain or schema server.

Input family	Fields
Epoch identity	Integer epoch ID and full GPST millisecond tick.
Initial state	ECEF (x, y, z) metres and receiver-clock metres.
Source operators	Independent hinge, three uint32 masks, explicit q, J, K , encoded three-bit blend.
Observation identity	Existing epoch ID, contiguous 0–31 slot and unique satellite identifier.
Prepared measurement	Satellite emission position in reception-time ECEF, corrected code metres and positive sigma metres.

The output includes the original epoch/tick, solver status, iteration count, selected count, all ASA fields, JK/blend outputs, four estimated state values, RMS, normalized residual sum, four formal sigmas, local ENU, log chart, OTAN2 fields, relation values/classes and both integer keys.

23.1 Numerical domain markers

Chart status zero means the compact key is defined. Status one means no solved position; two identifies the explicit local origin core; three denotes an out-of-chart radial or hinge value. A zero key with nonzero chart status is not a meaningful packed position.

OTAN2 status zero means the literal source ratio is defined. One means no eligible preceding sample; two is zero increment; three is a zero literal denominator; four is a numerical-range condition. In a pure-angular case the separate directed residual can still be defined. These fields are not silently unified into a fake zero.

23.2 Embedding

The shared library exposes `solve`, `enrich`, `cpu_run` and, in a CUDA build, `cuda_run`. A downstream application can use the estimated receiver position as data. This package contains no vehicle-control message encoder. Input preparation remains responsible for ephemerides, clock/propagation corrections, signal definitions and the common reception-time coordinate frame.

24 Package contents and conjoined coverage

Path	Delivered role
include/atomos/math.hpp	Shared geometry, coordinates, word primitives and exact key codecs.
include/atomos/nav.hpp	Weighted QR solve, typed results and conjoined enrichment.
src/	Prepared CSV interface, CPU loop and command-line executable.
cuda/backend.cu	Global/texture solve kernels, enrichment and bounded device allocation.
python/sclp.py	Independent finite UGTS helper implementation.
python/verify_result.py	Independent NumPy positioning/record oracle.
python/compare_backends.py	CPU/device field comparisons.
examples/	Clean and noisy/edge synthetic input batches and their known truth.
tests/, results/	Executed tests and actual numerical outputs.
tools/validate.py	CPU/device build and verification matrix.
docs/conjoined.tex	Editable source of this unified formalization.

24.1 What is source-derived and what is new

The aTOMos masks, JK/encoded blend, literal OTAN2 and its named completion are retained from the 3.6.1 project contract. The UGTS geometry, metric, topology separation, grammar/constraint relations and key layouts are derived from the supplied SCLP document and are identified by its section numbers.

SN1 adds the measurement model, weighted receiver-position/clock estimator, chosen local WGS84 anchor, prepared CSV schema and GPU mapping. Its numerical fixture parameters are labelled examples. The jitter-interval equality and nonnested sweep-density statement are explicitly distinguished where the printed source would otherwise be carried into an incorrect mathematical equality.

The supplied PDF describes an original UGTS archive with 153 prior tests. That archive was not supplied here; its count is not reported as a new result. This package's actual tests and CPU builds are the ones summarized in this volume. The earlier aTOMos master PDF likewise is retained through the project's stated contract, not presented as a newly mounted file that was compiled as source.

The delivered files require no provenance service. The following technical references identify the finite source contracts and the new navigation/API facts without adding a runtime ledger or historical claims chapter.

Technical references

UGTS. Tom Klootwijk, *UGTS–KC 3.6.2 — SCLP: Swept-Cone Log-Polar Packing*, the supplied 19-page PDF. The finite primitive and sweep are in §§3–4, chart calculus in §5, jitter/time in §§6–7, topology/hinge in §§8–9, grammar/refinement/keys in §§10–12 and the event handoff in §14. The source’s Table 1/Figure 6 supplies the exact 20/18/14/12 layout used here.

aTOMos. Tom Klootwijk, *aTOMos 3.6.1 — Total Integrated Engine*, the retained project specification of the ASA/NA word core, explicit JK/encoded blend, coherent records, literal OTAN2 and separately named completion. These operative equations are restated in this volume.

The following primary references support SN1 and the CUDA build contract. Consulted 14 September 2026.

- N1 ESA Navipedia, [Code Based Positioning \(SPS\)](#). Corrected-code four-unknown positioning and linearization.
- N2 ESA Navipedia, [Satellite Coordinates Computation](#). Emission coordinates in a common reception-time ECEF frame.
- N3 ESA Navipedia, [Weighted Least Square Solution](#). Weighted objective and formal covariance interpretation.
- N4 ESA Navipedia, [Transformations between ECEF and ENU coordinates](#). Local orthonormal frame.
- N5 NGA Office of Geomatics, [World Geodetic System 1984](#). Ellipsoid defining parameters.
- H1 NVIDIA, [GeForce RTX 50 Series Laptops](#), memory/specifications table.
- H2 NVIDIA, [CUDA GPU Compute Capability](#), RTX 5070 Ti family.
- H3 NVIDIA, [CUDA 12.8 Features Archive](#), introduction of SM_120 compiler support.
- H4 NVIDIA, [CUDA Runtime API: Texture Object Management](#).

aTOMos v3.6.1.5 / UGTS Conjoined Satnav Kernel.

Tom Klootwijk | NL200678942 | 10-07-1990.

Unified mathematical specification, editable C++/CUDA source and recorded local verification.